Course Code: ECE1003	**Course Title:** Digital Logic Design	**TPC**	**3**	**2**	**3**
Version No.	**1.1**				
Course Pre-requisites/ Co-requisites/ anti-requisites (if any).	**None**				
Objectives:	1. To learn various techniques for logic minimization. 2. To comprehend the concepts of various combinational circuits. 3. To understand the concepts of various sequential circuits. 4. Helps student to carry out projects which contain digital electronics modules.				

CO's Mapping with PO's and PEO's

Course Outcomes	Course Outcome Statement	PO's / PEO's
CO1	Assimilate the knowledge of different types of number systems and logically use them to perform binary arithmetic	PO1, PO2, PO3, PEO1
CO2	Design and Analyze combinational and sequential circuits.	PO1, PO2, PO3, PEO1, PEO2
CO3	Design and analyze finite state machines	PO1, PO2, PO3, PEO1, PEO2
CO4	Design and realize combinational and sequential circuits using programmable devices	PO1, PO2, PO3, PEO1, PEO2
CO5	Assess different logic families that exist to realize digital circuits and understand their operation and limitations	PO1, PO2, PO3, PEO1, PEO1
		TOTAL HOURS OF INSTRUCTIONS: 44

Module No. 1	**Number Systems and Boolean Algebra**	**12 Hours**
Number systems and conversions, r's and (r-1)'s compliment, binary signed and unsigned numbers, binary arithmetic operations, weighted and non-weighted binary codes, logic gates and universal gates, tri-state logic and don't care logic, boolean algebra, SOP and POS minimization, K-maps upto 4-variables.		
Module No. 2	**Combinational Logic Circuit Design**	**12 Hours**
Binary adder and subtractor, Ripple carry adders/subtractors and fast adders. Binary decoders, encoders, multiplexers and de-multiplexers. Logic functions using decoders and multiplexers. , Code converters, Magnitude comparator.		
Module No. 3	**Sequential Logic Circuit Design**	**13Hours**
Latches, Flip flops, Flip flop conversions, shift Registers, counters-synchronous and asynchronous counters, ring and Johnson counters. Finite State Machine – design using mealy and Moore state machines, Sequence detectors and generators design.		
Module No. 4	**Programmable Logic Devices**	**5 Hours**
PROM, PLA, PAL, Comparison between PROM, PLA and PAL. Look-Up Tables (LUT) design.		
Module No. 5	**Digital Integrated Circuits**	**6 Hours**

Introduction to various logic families, Specifications – noise margin, propagation delay, fan-in, fan-out. Comparison of logic families. Basic combination circuits design using NMOS, PMOS and CMOS logic families.	
Text Books. 1. M.Morris Mano, Michael D Ciletti, Digital Design , 5th edition, Pearson Publishers, 2013. 2. Samir Palnitkar, "Verilog HDL: A Guide to Digital Design and Synthesis" Prentice Hall, 2ndEdition, 2009. 3. R.P. Jain, "Modern Digital Electronics", 4th edition,TMH.	
References 1. M.Morris Mano, Charles R. Kime, Tom MartinLogic and Computer Design Fundamentals, 4thedition, Pearson Publishers. 2. C. H. Roth and L. L. Kinney, Fundamentals of Logic Design, 5th edition, Cengage Publishers	
Mode of Evaluation	
Recommended by the Board of Studies on	07.04.2019
Date of Approval by the Academic Council	Xxth Academic Council held on xx-xx-xxxx

List of Laboratory Experiments

1. Verification of Basic gates in behavioral, dataflow and gate-level modeling.
2. Verification of half adder & subtractor, full adder and subtractors.
3. Design and verification of 4-bit binary adder and subtractor.
4. Design and verification of 3x8 decoder and 8x3 priority encoder.
5. Design and verification of 4x1 MUX and 1x4 DeMUX.
6. Design and verification of 4-bit magnitude comparator.
7. Verification of latches and flip-flops.
8. Design and verification of 4-bit shift registers – SISO, SIPO, PISO and PIPO
9. Design and verification of 4-bit asynchronous up, down, up/down counter.
10. Design and verification of 4-bit synchronous up, down, up/down counter.
11. Design and verification of 4-bit ring and Johnson counters.